

Numerical modelling of soil water plant interaction

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ABSTRACT: The mathematical modelling of transient soil water flow in the presence of plant roots has been undertaken by several authors. In most of the models, the water uptake by roots is represented by a volumetric sink term, which is added to the continuity equation for soil water flow in the unsaturated zone. The sink term in the model needs detailed information about the root system such as root radius, root length, root density etc. Experimental evaluation of root system is both time consuming and costly. Therefore, to overcome this, the present paper deals with the solution of transient two dimensional axi-symmetric radial flow Richards equation where the volumetric sink term is computed by assigning known suction head at the root-soil interface as a boundary condition. The governing Richards equation is solved numerically by using an Iterative Alternating Direction Implicit explicit (IADIE) finite difference scheme.

1 INTRODUCTION

The availability of water for plant roots is an important topic which has been explored by a number of investigators, e.g. Gardner, (1960,1964), Whisler et al., (1968), Molz and Remson, (1970, 1971,1973), Belmans et al., (1979), de Jong and Cameron , (1979), McCoy et al., (1984). The research has followed basically in two different approaches: Microscopic approach and Macroscopic approach. In microscopic approach radial moisture flow towards a single root was analysed (Philip,1957; Gardner, 1960 and Molz et al.,1968). In macroscopic approach (Gardner, (1964), Whisler et al. (1968), and Feddes et al., (1974)) the attention is focussed on the removal of moisture from a differential volume of a soil as a whole without considering the effect of individual roots. In such an approach, the removal of moisture from roots is represented by a volumetric sink term. The sink term in these models requires detailed information of root characteristics. Molz et al.(1970) and Nimah et al.(1973a) have incorporated the sink term while modelling the water uptake by plants with some assumptions. However, these approximations may not accurately describe the root water uptake. Feddes et al. (1978) and Hoogland et al. (1981) have given more simpler and practical functions for the estimation of sink term. The numerical solution of

Richard's equation with sink term requires definition of an additional constituent parameter in the form of sink term. The sink term is described by a geometrical shape which is a function of soil water content at the field capacity, permanent wilting point, change in moisture storage and root characteristics.

To overcome this prior knowledge of sink term for the solution of Richard's equation, a suction head at the root-soil interface as a boundary condition is assigned because it is easier to know the suction head at the root-soil interface rather than the experimental evaluation of root characteristics and prior knowledge of the variation of soil moisture content in the root zone. The model requires only the information of depth of root length and the root radius.

The objective of this study is to develop a mathematical model for the water uptake by plant and to use it for some field data. Here, uniform root growth is considered

2 MODEL DEVELOPMENT

To model flow towards root in unsaturated zone, suction head based two dimensional axi-symmetric nonlinear parabolic differential Richard's equation is preferred which is given by:

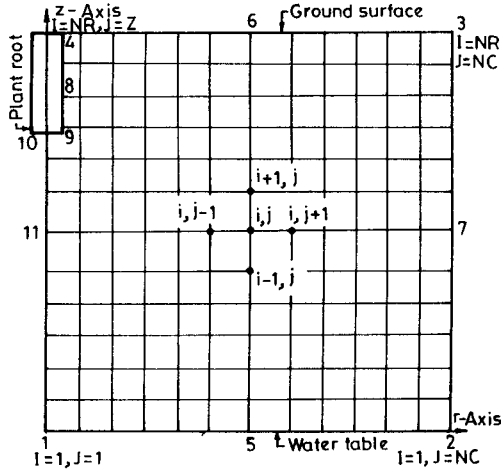


Fig. 1 Finite Difference Mesh & Boundaries

$$C \frac{\partial h}{\partial t} = \frac{1}{r} \frac{\partial}{\partial r} \left[r K_r \frac{\partial (-h)}{\partial r} \right] + \frac{\partial}{\partial z} \left[K_z \frac{\partial (-h+z)}{\partial z} \right] \quad (1)$$

Where, r is the distance outward from the main plant axis, z is vertical coordinate in the upward direction, h is suction head, K_r and K_z are unsaturated hydraulic conductivities in the r and z direction respectively, $C (=d\theta/dh)$ is specific moisture capacity and t is the time.

Referring to Figure 1, the initial and boundary conditions are as follows:

Initial condition

$$(a) \theta(z, r, 0) = \theta_0(z), \quad t=0, 0 \leq z \leq L, 0 < r < R$$

Where L is the depth of the domain in the z -direction and R is the width of the domain in r direction.

Boundary conditions

(b) Boundary 1-2 (Lower boundary)

Water table is considered as the lower boundary condition i.e.,

$$h = 0, t > 0, z = 0, 0 < r < R$$

(c) Boundary 2-3 (Far end constant head boundary)

Here Dirichlet type boundary condition is assigned:

$$h = HH, t > 0, 0 < z < L, r = R$$

where, HH is the initial suction head.

(d) Boundary 3-4 (Upper boundary)

Infiltration due to either rainfall or applied irrigation is considered as the upper boundary condition, which is as follows:

$$q = k_z \left(\frac{\partial h}{\partial z} - 1 \right), t > 0, z = L, 0 < r < R$$

(e) Boundary 4-9 (Root-Soil interface)

constant suction head at root-soil interface is assigned as follows:

$$h(z, r) = H_{\text{root}}; \quad 0 \leq z \leq \text{root length at all times.}$$

(f) Boundary 1-9 (Water divide)

Boundary 1-9 is the vertical axis extending from the centre of the bottom centre of the root to the ground water table. As the flow is axis symmetric, this can be considered as a water divide and hence no flow across the boundary. Neuman type boundary condition is assigned with $q=0$ as follows:

$$\frac{\partial h}{\partial r} = 0.0 \quad \text{at } (z, 0); \quad 0 \leq z \leq C \quad \text{at all times.}$$

Where C is the depth between bottom of the root to the ground water.

3 NUMERICAL SOLUTION

Equation (1) in terms of finite difference, subject to the initial and boundary conditions (a)-(f), is solved by the Iterative Alternating Direction Implicit Explicit (IADIE) method (Remson et al.1971; Ahmed et al.1990,1991) proposed by Peacemen and Rachford(1955), where the flow domain is discretised by a finite number of nodal points. The time domain is also discretised by a finite number of discrete time steps. The soil characteristic data (θ , K_r , K_z and C) for the unsaturated zone are generated within the model, in accordance with the following equation:

3.1 K - Characteristic

The following relations, proposed by Brooks and Corey(1964) are adopted:

$$K(r, \theta) = K_{\text{sat}} \left(\frac{\theta - \theta_r}{\phi - \theta_r} \right)^\lambda; \quad \theta \geq \theta_r \quad (2)$$

$$K(r, \theta) = 0.0 ; \theta < \theta_r \quad (3)$$

$$K(z, \theta) = K_{sat} \left(\frac{\theta - \theta_r}{\phi - \theta_r} \right)^\lambda ; \theta \geq \theta_r \quad (4)$$

$$K(z, \theta) = 0.0 ; \theta < \theta_r \quad (5)$$

Where, θ_r is residual moisture content; ϕ is porosity; K_{sat} is hydraulic conductivity at saturation; θ is volumetric moisture content and λ is pore size index.

3.2 θ - Characteristic

The following relation (Rao et al. 1986) is adopted:

$$\theta = \exp \left[\frac{\ln(\phi - 2\theta_r)}{h_b} h \right] + \theta_r ; h \geq h_b \quad (6)$$

$$\theta = \phi - \frac{\theta_r}{h_b} h ; 0 \leq h \leq h_b \quad (7)$$

Where, h is matric suction head; h_b is air entry value (bubbling pressure) and other variables are defined earlier.

The solution obtained in the form of suction head at different times at all nodes of the domain by IADIE method is again solved by Picard's iteration scheme.

3.3 Convergence criteria

At the end of a discrete time step the difference of h values of successive iteration at each of the finite difference nodes is computed. The IADIE iteration are continued until the biggest of these differences is less than a pre-stipulated convergence factor ϵ . Thus, for IADIE as well as Picard's solution, the convergence criteria is as follows:

$$\text{Max}[Abs(h_i - h_{i-1})] < \epsilon \quad (8)$$

Where h_i = the suction head at a node after i^{th} IADIE iteration, h_{i-1} = the suction head at the node after $(i-1)^{\text{th}}$ iteration.

4 MODEL APPLICATION

The data for the initial soil moisture content at the

time of crop sowing and soil physical properties like porosity, saturated hydraulic conductivity etc. are observed at the demonstration farm of Water Resources Development Training Centre, University of Roorkee, Roorkee (India) during Rabi(winter) 1990-1991.

The initial soil moisture at the time of sowing are given below:

Soil depth(cm): 00-30 30-60 60-90 90-120 120-150

Soil moisture : 0.2451 0.2552 0.2650 0.2751 0.2848 (cc/cc)

Soil properties (physical): Bulk density = 1.36 g/cc; Particle density = 2.53g/cc; Porosity (%) = 0.462; Saturated hydraulic conductivity = 0.5517mm/hr; Field capacity = 21.00(g/g %); Permanent wilting point = 13.50g/g%; Bubbling pressure (estimated) = 655.62mm ; Pore size index (λ) (estimated) = 4.0

Plant properties:

The root radius and its length are observed as 16mm and 200mm after the initial crop stage (26 days). 30mm irrigation on 22nd day and 15mm irrigation on 24th day of sowing is applied.

The suction head at the root-soil interface which depends upon the atmospheric demand and type of the crop, is taken as 0.2 bar for the wheat crop (initial stage) from available published data.

The proposed numerical model simulates the above observed data in the form of moisture content and suction head distribution at the root zone depth with respect to time. Figure 2 shows the contour of moisture content at the root zone depth with respect to time. Figure 3 shows the variation of amount of water uptake by plant with respect to time and compared with field estimated data.

5 CONCLUSIONS

The numerically simulated moisture content distributions at the root zone depth with respect to time would help in the estimation of the amount of water uptake by the plant if free flow movement of water is considered from root surface to the top of the plant leaves to the atmosphere which leads to the computation of the transpiration rate. This approach is easier and numerically suitable for the estimation of transpiration rate.

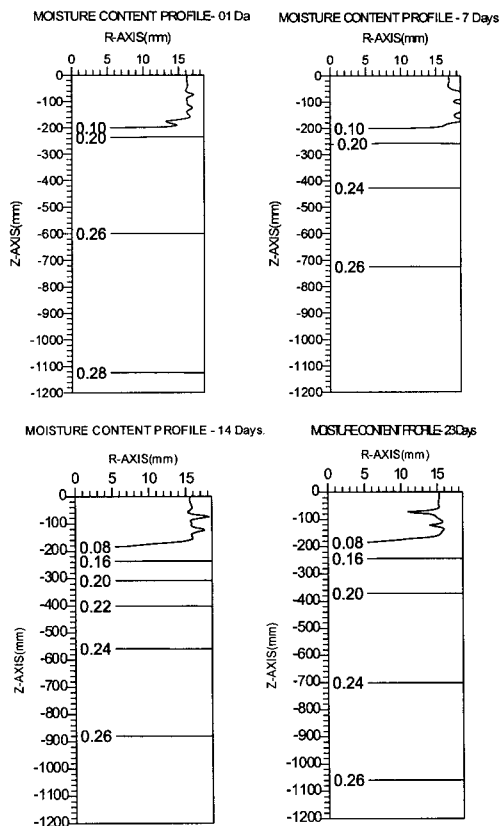


Figure 2 Contours of moisture content on 1,7,14 and 23 days.

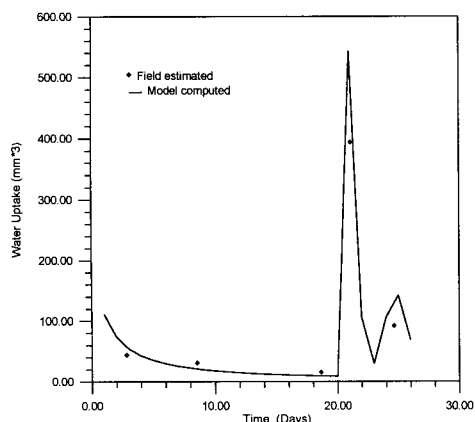


Figure 3 Water uptake versus time.

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